

Southern University of Science and Technology
EE205 Signals and Systems Spring 2023 Midterm Exam

Last Name: _____

First Name: _____

Your ID: _____

INSTRUCTIONS:

- Please do not sit next to each other.
- This exam is closed books and closed notes.
- You are only allowed to bring with you a two-sided sheet of A4. In order for you to get your midterm grade, you need to turn in your note with your exam. Please write down your name on the note.
- Time limit: 2:00 – 3:50 PM.
- Please print your name clearly at the location indicated.

Problem	Points	Notes
1		
2		
3		
4		
5		
6		
7		Extra points
Total		

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1. (15 points) A system is described by

$$\frac{dy(t)}{dt} = 3\frac{dx(t)}{dt} + t^2x(t).$$

Show that whether or not the system is linear.

The system is linear.

Let us use $x(t) \rightarrow y(t)$ to described the system

$$\frac{dy(t)}{dt} = 3\frac{dx(t)}{dt} + t^2x(t).$$

That is to say, if $x_1(t) \rightarrow y_1(t)$, then

$$\frac{dy_1(t)}{dt} = 3\frac{dx_1(t)}{dt} + t^2x_1(t) \tag{A}$$

and if $x_2(t) \rightarrow y_2(t)$, then

$$\frac{dy_2(t)}{dt} = 3\frac{dx_2(t)}{dt} + t^2x_2(t). \tag{B}$$

From $\alpha(A) + \beta(B)$, we have

$$\frac{d[\alpha y_1(t) + \beta y_2(t)]}{dt} = 3\frac{d[\alpha x_1(t) + \beta x_2(t)]}{dt} + t^2[\alpha x_1(t) + \beta x_2(t)].$$

Clearly, we have

$$\alpha x_1(t) + \beta x_2(t) \rightarrow \alpha y_1(t) + \beta y_2(t)$$

and therefore, the system is linear.

2. Determine whether or not each of the following signals is periodic. If the signal is periodic, determine its fundamental period.

(a) (10 points)

$$x[n] = \sin\left(\frac{7\pi}{6}n + 2\right).$$

(b) (5 points)

$$x[n] = \sin\left(\frac{\pi}{7}n^2\right).$$

(a) *This signal is periodic. For an integer N , we need*

$$\frac{7\pi}{6}N = 2m\pi$$

where m is an integer. We have that the smallest N can be obtained when $m = 7$.

The period is $N = 12$.

(b) *This signal is periodic. For an integer N , we need*

$$\frac{\pi}{7}(n + N)^2 = \frac{\pi}{7}(n^2 + 2nN + N^2) = \frac{\pi}{7}n^2 + \frac{\pi}{7}(2nN + N^2) = \frac{\pi}{7}n^2 + 2m\pi$$

where m is an integer. We have that the smallest N that satisfies the above condition is $N = 14$.

Therefore, the period is $N = 14$.

3. (15 points) A continuous-time signal $x(t)$ is known to be periodic with period $T = 4$. In its Fourier series representation, there are only four nonzero coefficients and they are given by

$$a_1 = \frac{j}{2}, \quad a_{-1} = -\frac{j}{2}, \quad a_4 = a_{-4} = 1.$$

Show that $x(t)$ is a real function by determining an expression for $x(t)$.

From Fourier series representation, we have

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = a_1 e^{j\omega_0 t} + a_{-1} e^{-j\omega_0 t} + a_4 e^{j4\omega_0 t} + a_{-4} e^{-j4\omega_0 t},$$

where $\omega_0 = 2\pi/T = \pi/2$.

We therefore have

$$\begin{aligned} x(t) &= \frac{j}{2} e^{j(\pi/2)t} - \frac{j}{2} e^{-j(\pi/2)t} + e^{j4(\pi/2)t} + e^{-j4(\pi/2)t} \\ &= -\frac{e^{j\pi t/2} - e^{-j\pi t/2}}{2j} + 2 \cos 2\pi t = -\sin \frac{\pi}{2}t + 2 \cos 2\pi t, \end{aligned}$$

and $x(t)$ is a real function.

4. (15 points) Compute and plot $y[n] = x[n] * h[n]$, where

$$x[n] = \begin{cases} 1, & 2 \leq n \leq 5, \\ 0, & \text{otherwise,} \end{cases}$$

$$h[n] = \begin{cases} 1, & n = 1, \\ 2, & n = 2, \\ 1, & n = 3, \\ 0, & \text{otherwise,} \end{cases}$$

By definition, we have

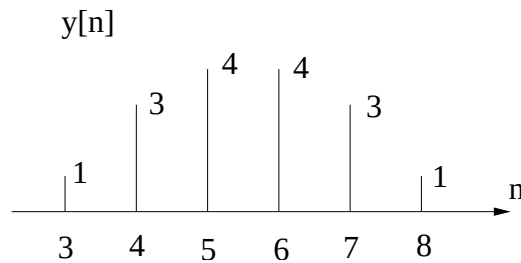
$$\begin{aligned} y[n] &= x[n] * h[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = \sum_{k=1}^3 h[k]x[n-k] \\ &= h[1]x[n-1] + h[2]x[n-2] + h[3]x[n-3]. \end{aligned}$$

That is to say

$$y[n] = x[n-1] + 2x[n-2] + x[n-3].$$

Therefore,

$$y[n] = \begin{cases} 1, & n = 3, \\ 3, & n = 4, \\ 4, & n = 5, \\ 4, & n = 6, \\ 3, & n = 7, \\ 1, & n = 8, \\ 0, & \text{otherwise.} \end{cases}$$



5. (20 points) Use the definition of Fourier transform to calculate the Fourier transform $X(j\omega)$ of $x(t) = e^{-3t}u(t-2)$. Also, determine $|X(j\omega)|$ and $\angle X(j\omega)$.

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt = \int_2^{\infty} e^{-3t}e^{-j\omega t} dt = \frac{-1}{3+j\omega} e^{-(3+j\omega)t} \Big|_2^{\infty} \\ &= \frac{e^{-(3+j\omega)2}}{3+j\omega}. \end{aligned}$$

$$|X(j\omega)| = \frac{e^{-6}}{\sqrt{9+\omega^2}}$$

and

$$\angle X(j\omega) = -2\omega - \arctan \frac{\omega}{3}.$$

6. Consider the signal given by

$$x(t) = \begin{cases} e^{-t}, & 0 \leq t \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Two other signals are given by

$$y(t) = \begin{cases} e^{-t}, & 0 \leq t \leq 1, \\ e^t, & -1 \leq t \leq 0, \\ 0, & \text{otherwise,} \end{cases}$$

$$z(t) = \begin{cases} e^{-t}, & 0 \leq t \leq 1, \\ -e^t, & -1 \leq t < 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) (10 points) Determine the Fourier transform $X(j\omega)$ of $x(t)$.

(b) (10 points) Use Fourier transform properties to determine the Fourier transforms $Y(j\omega)$ and $Z(j\omega)$ of $y(t)$ and $z(t)$, respectively, from $X(j\omega)$. Show that $Y(j\omega)$ is real and $Z(j\omega)$ is imaginary.

(a) The Fourier transform of $x(t)$ is given by

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt = \int_0^1 e^{-t} e^{-j\omega t} dt = -\frac{1}{1+j\omega} e^{-(1+j\omega)t} \Big|_0^1 = \frac{1}{1+j\omega} (1 - e^{-(1+j\omega)}) \\ &= \frac{1 - e^{-1} e^{-j\omega}}{1+j\omega} = \frac{1 - e^{-1} \cos \omega + j e^{-1} \sin \omega}{1+j\omega}. \end{aligned}$$

(b) We can express

$$y(t) = x(-t) + x(t), \quad z(t) = -x(-t) + x(t).$$

Therefore,

$$Y(j\omega) = X(-j\omega) + X(j\omega) = 2\operatorname{Re}\{X(j\omega)\} = 2\frac{1 - e^{-1} \cos \omega + \omega e^{-1} \sin \omega}{1 + \omega^2} \text{ is real,}$$

$$Z(j\omega) = -X(-j\omega) + X(j\omega) = 2j\operatorname{Im}\{X(j\omega)\} = 2j\frac{e^{-1} \sin \omega - \omega(1 - e^{-1} \cos \omega)}{1 + \omega^2} \text{ is imaginary.}$$

7. (10 points – Extra credit) Determine the period and the Fourier transform of the periodic signal $x(t) = \sin \frac{\pi}{3}t + \cos \frac{\pi}{2}t$.

The period of $x(t)$ is $T = 12$.

We have $\omega_0 = \frac{2\pi}{T} = \frac{\pi}{6}$.

$x(t)$ can be expressed as

$$\begin{aligned} x(t) &= \sin \frac{\pi}{3}t + \cos \frac{\pi}{2}t = \sin 2\omega_0 t + \cos 3\omega_0 t \\ &= \frac{e^{j2\omega_0 t} - e^{-j2\omega_0 t}}{2j} + \frac{e^{j3\omega_0 t} + e^{-j3\omega_0 t}}{2}. \end{aligned}$$

From $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$, we obtain

$$a_{-3} = \frac{1}{2}, \quad a_3 = \frac{1}{2}, \quad a_{-2} = -\frac{1}{2j}, \quad a_2 = \frac{1}{2j}.$$

Then, for the Fourier transform of a periodic signal, we have

$$\begin{aligned} X(j\omega) &= \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0) \\ &= 2\pi a_{-3} \delta(\omega + 3\omega_0) + 2\pi a_{-2} \delta(\omega + 2\omega_0) + 2\pi a_2 \delta(\omega - 2\omega_0) + 2\pi a_3 \delta(\omega - 3\omega_0) \\ &= \pi \delta\left(\omega + \frac{\pi}{2}\right) - \frac{\pi}{j} \delta\left(\omega + \frac{\pi}{3}\right) + \frac{\pi}{j} \delta\left(\omega - \frac{\pi}{3}\right) + \pi \delta\left(\omega - \frac{\pi}{2}\right). \end{aligned}$$